

# Week 1: Circuit Concepts

Charge, current, voltage, power, KCL, KVL, series/parallel, dividers, equivalent resistance

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 17 worked examples, and 8 practice problems with full solutions.

**Based on:** Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §1.1–1.6 and §2.1–2.3

**Inspired by:** ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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# How to use this deck

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- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** 8 final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

# Symbols and units

| Symbol | Name        | Unit             | Plain meaning                               |
|--------|-------------|------------------|---------------------------------------------|
| $q$    | charge      | coulomb (C)      | amount of electricity                       |
| $i, I$ | current     | ampere (A)       | charge flowing past a point per second      |
| $v, V$ | voltage     | volt (V)         | energy per charge <i>between two points</i> |
| $R$    | resistance  | ohm ( $\Omega$ ) | how much it blocks current                  |
| $G$    | conductance | siemens (S)      | $1/R$ , how easily current flows            |
| $p, P$ | power       | watt (W)         | energy per second                           |
| $w, W$ | energy      | joule (J)        | total work done                             |

Lowercase  $v, i$  = can change with time. Uppercase  $V, I$  = constant (DC). Weeks 1–3 are all DC.

# Charge, current, voltage, energy

[NOT on formula sheet: memorize]

Current  $i = dq/dt$       constant current:  $q = I t$

Voltage  $v = dw/dq$       energy per unit charge

Power  $p = dw/dt = v i$

Energy  $w = \int p dt$       constant power:  $W = P t$

1 A = 1 C/s    1 V = 1 J/C    1 W = 1 J/s

1 A·h = 3600 C    1 kWh = 3.6 MJ

## Trap

A·h (amp-hour) is **charge**, not energy. Energy = voltage  $\times$  charge.

## Units trick: $\text{k}\Omega \times \text{mA} = \text{V}$

Engineering shortcut: kilo-ohms and milliamps cancel

$$\text{k}\Omega \times \text{mA} = \text{V}$$

$$\text{V} \div \text{k}\Omega = \text{mA}$$

Example:  $5 \text{ V} / 2 \text{ k}\Omega = 2.5 \text{ mA}$ . No  $10^3$  or  $10^{-3}$  needed.

$$\text{m} = 10^{-3} \quad \text{k} = 10^3 \quad \mu = 10^{-6} \quad \text{M} = 10^6$$

### Trap

$\text{mA} \times \Omega$  gives **mV**, not V. Exams often mix  $\Omega$  and  $\text{k}\Omega$  in one circuit, so convert everything to the same unit first.

## Try it: Example 1, units

### Example 1

- (a) 12 V across 4 k $\Omega$ . Current?
- (b) 2.5 mA through 2 k $\Omega$ . Voltage?
- (c) A 12 V battery is rated 60 A·h. How much charge (C) and energy (J and Wh) does it hold?
- (d) A 100 W lamp runs for 5 h. Energy in kWh and in J?

your work here ↓

# Example 1: solution

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(a)  $12 \text{ V} / 4 \text{ k}\Omega = 3 \text{ mA}$

(b)  $2.5 \text{ mA} \times 2 \text{ k}\Omega = 5 \text{ V}$

(c) **charge**  $60 \text{ A} \times 3600 \text{ s} = 216\,000 \text{ C}$

(c) **energy**  $W = Vq = 12 \times 216\,000 = 2.592 \text{ MJ}$

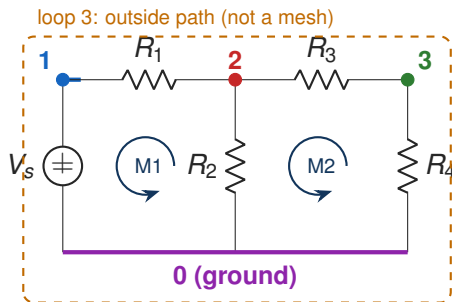
or in Wh:  $12 \text{ V} \times 60 \text{ A}\cdot\text{h} = 720 \text{ Wh}$

(d)  $W = Pt = 100 \text{ W} \times 5 \text{ h} = 500 \text{ Wh} = 0.5 \text{ kWh}$

in joules:  $0.5 \times 3.6 \text{ MJ} = 1.8 \text{ MJ}$

**Shortcut:** Wh  $\rightarrow$  J and A·h  $\rightarrow$  C: both multiply by 3600.

# Nodes, branches, loops, meshes



**Branch:** one element. Here: 5

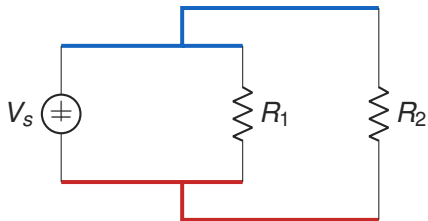
**Node:** a connection point (colored dots). Here: 4

**Loop:** any closed path. Here: 3

**Mesh:** a loop with nothing inside it, like a window. Here: M1, M2

Loop 3 goes around the outside through  $V_s$ ,  $R_1$ ,  $R_3$ ,  $R_4$ . It contains M1 and M2, so it's **not** a mesh.

# Most important rule: one wire = one node



All blue wire is **one node**. All red wire is **one node**.

Only **2 nodes**, so  $V_s$ ,  $R_1$ ,  $R_2$  all have the same voltage.

Wire length and shape never matter.

## Ideal wire and ground

Ideal wire: **no voltage drop**. Ground symbol: the **reference node** (0 V). Every ground symbol is the **same node**.

# Counting check: meshes = $b - n + 1$

## Engineering shortcut: count check

For a flat circuit:  $\# \text{ meshes} = b - n + 1$

Two slides back:  $b = 5$ ,  $n = 4$ , so meshes =  $5 - 4 + 1 = 2$  ✓

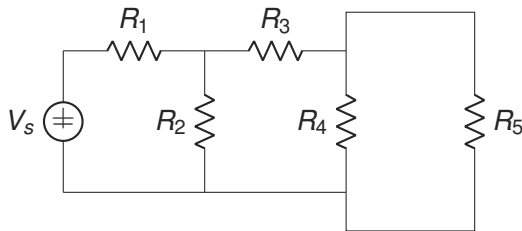
If it doesn't check out, you probably split one node into two.

Week 2 preview: mesh analysis needs 1 equation per mesh, and nodal needs  $n - 1$ .

## Rule

$N$  unknowns need  $N$  **independent** equations. Using meshes guarantees that.

## Try it: Example 2, count the circuit



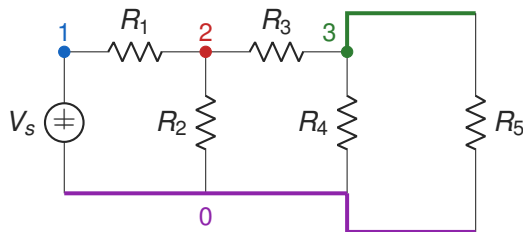
Count:

- (a) nodes
- (b) branches
- (c) meshes

Check with  $b - n + 1$ .

your work here ↓

## Example 2: solution



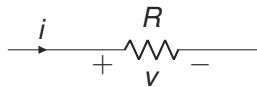
|                 |                              |
|-----------------|------------------------------|
| <b>Nodes</b>    | 4 (1, 2, 3, 0)               |
| <b>Branches</b> | 6 ( $V_s$ , $R_1$ to $R_5$ ) |
| <b>Meshes</b>   | 3                            |
| <b>Check</b>    | $6 - 4 + 1 = 3 \checkmark$   |

### Trap

The long green wire makes the top of  $R_5$  look like a new node. It isn't: it's node 3. So  $R_4$  and  $R_5$  are in **parallel**.

# Resistor, Ohm's law, resistivity

[on formula sheet]



$$v = iR \quad p = vi = i^2 R = \frac{v^2}{R}$$

- $v = iR$  only when the current arrow points **into the + side**. Otherwise  $v = -iR$ .
- A resistor **always absorbs** power. Negative resistor power means a sign mistake.
- Conductance:  $G = 1/R$  (siemens).
- Resistivity:  $R = \rho \ell / A$ . Longer wire means more  $R$ , thicker means less.

# Short circuit vs open circuit

wire



**Short circuit**

$$R = 0$$

voltage across it = 0

any current can flow

gap



**Open circuit**

$$R = \infty$$

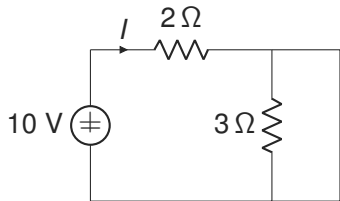
current through it = 0

any voltage can appear

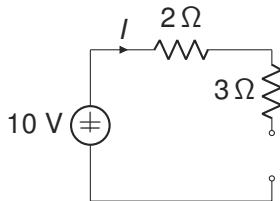
## Engineering shortcut

Short  $\Rightarrow$  write " $v = 0$ ".    Open  $\Rightarrow$  write " $i = 0$ ". Each one hands you a free equation.

## Try it: Example 3, short and open



(a) wire across the  $3\ \Omega$



(b) a gap under the  $3\ \Omega$

For each: find  $I$ , the voltage across the  $3\ \Omega$ , and (b) the voltage across the gap.

your work here ↓

## Example 3: solution

(a) wire  $\parallel 3\ \Omega$ :  $V_{3\Omega} = 0\ \text{V}$ , and no current in the  $3\ \Omega$

only the  $2\ \Omega$  is left:  $I = 10/2 = 5\ \text{A}$

(b) the gap is in series with everything:  $I = 0\ \text{A}$

no current means no drops:  $V_{3\Omega} = 0\ \text{V}$

KVL: all  $10\ \text{V}$  appears across the gap:  $10\ \text{V}$

### Trap

An open circuit is not “ $0\ \text{V}$ ”. It is “ $0\ \text{A}$ ”. The voltage across a gap can be large.

# Ideal sources: what's fixed, what's free



## Voltage source

Fixed: the **voltage** across it

Free: its **current** (+, −, or 0)

Killed (set to 0) → **short**



## Current source

Fixed: the **current** through it

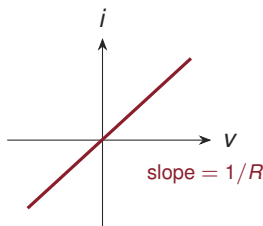
Free: its **voltage** (you solve for it)

Killed (set to 0) → **open**

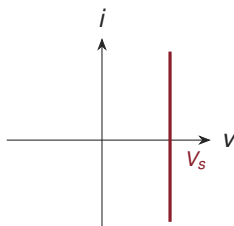
### Common exam question

“Find the voltage across the current source”: the source doesn’t tell you. Use KVL around a loop that contains it (Example 4).

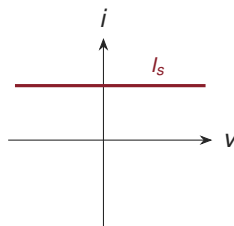
# $i-v$ graphs of the three basic elements



**Resistor**



**Voltage source**



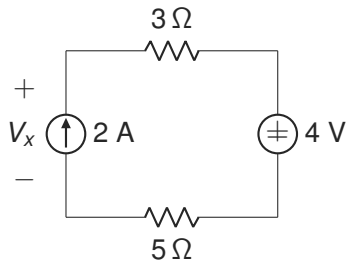
**Current source**

Voltage source: the voltage is always  $V_s$ , whatever the current (vertical line).

Current source: the current is always  $I_s$ , whatever the voltage (horizontal line).

Resistor: a straight line through the origin, which is Ohm's law.

## Try it: Example 4, voltage across an I source



Find  $V_x$  (top is  $+$ ) and the power of the current source. Check the power balance.

your work here ↓

## Example 4: solution

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**Current** one loop: everything carries 2 A clockwise

**KVL**  $V_x = \text{sum of drops around the loop: } 3(2) + 4 + 5(2)$

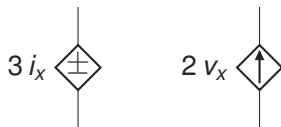
**Result**  $V_x = 20 \text{ V}$

**/ source** 2 A leaves its +:  $p = -(20)(2) \Rightarrow$  supplies 40 W

**Absorbed**  $3 \Omega$ : 12 W,  $5 \Omega$ : 20 W, 4 V source: 8 W

**Check**  $12 + 20 + 8 = 40 \checkmark$

# Dependent (controlled) sources



diamond = dependent

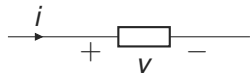
|      | Output  | Depends on    | Value                |
|------|---------|---------------|----------------------|
| VCVS | voltage | voltage $v_x$ | $\mu v_x$            |
| CCVS | voltage | current $i_x$ | $r i_x$ ( $\Omega$ ) |
| VCCS | current | voltage $v_x$ | $g v_x$ (S)          |
| CCCS | current | current $i_x$ | $\beta i_x$          |

## The two rules

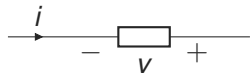
1. In KCL/KVL, treat it like a normal source whose value is an expression. Add one equation that links  $i_x$  or  $v_x$  to your unknowns.
2. It is **never killed** when finding  $R_{eq}$  or  $R_{Th}$  (week 3). Only independent sources get killed.

# Power: the passive sign convention

[NOT on formula sheet: memorize]



$i$  enters  $+$   $\Rightarrow p = +vi$



$i$  enters  $-$   $\Rightarrow p = -vi$

$p$  = power **absorbed**

$p > 0 \Rightarrow$  **absorbs**

$p < 0 \Rightarrow$  **supplies**

## Engineering shortcut

Always use the rule, then let the sign say absorb or supply. Never guess from “it’s a source”.

# Power balance, and sources that absorb

## Conservation of energy

$$\sum p_{\text{absorbed}} = 0 \quad \Longleftrightarrow \quad \sum p_{\text{supplied}} = \sum p_{\text{absorbed}}$$

## Free answer check

On any “find the power” question, add everything up. If it doesn’t balance, something is wrong. It takes 20 seconds.

## Sources don’t always supply

A source can **absorb** power (a battery being charged). Exams often ask you to “identify the source(s) of power” to test exactly this.

## Try it: Example 5, passive sign convention

Find the power absorbed. Absorbing or supplying?

(a)  $v = 10 \text{ V}$ ,  $i = 2 \text{ A}$  into  $+$       (b)  $v = 10 \text{ V}$ ,  $i = 2 \text{ A}$  into  $-$

(c)  $v = -5 \text{ V}$ ,  $i = 3 \text{ A}$  into  $+$       (d)  $v = -5 \text{ V}$ ,  $i = -3 \text{ A}$  into  $+$

your work here ↓

## Example 5: solution

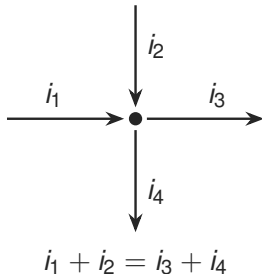
|     | Current enters | Formula                              | Result        |
|-----|----------------|--------------------------------------|---------------|
| (a) | +              | $p = vi = (10)(2) = +20 \text{ W}$   | absorbs 20 W  |
| (b) | -              | $p = -vi = -(10)(2) = -20 \text{ W}$ | supplies 20 W |
| (c) | +              | $p = vi = (-5)(3) = -15 \text{ W}$   | supplies 15 W |
| (d) | +              | $p = vi = (-5)(-3) = +15 \text{ W}$  | absorbs 15 W  |

### Why (c) works

A negative voltage means the real + is on the other terminal. Don't redraw anything. Plug in the signs and the algebra handles it.

# KCL: current in = current out

[on formula sheet]



$$\sum i_{\text{in}} = \sum i_{\text{out}}$$

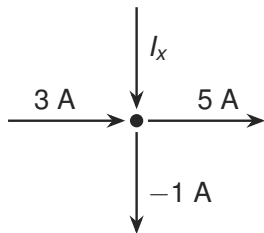
at **every node** (conservation of charge).

Sheet form:  $\sum i_n = 0$  (in +, out -).

## Engineering shortcut: KCL on a whole region

KCL works on any closed boundary around part of a circuit (a **supernode**):  
current in = current out, without solving inside.

## Try it: Example 6, KCL



(a)

Find  $I_x$  and  $I_y$ .



(b) unknown circuit inside

your work here ↓

## Example 6: solution

(a) in = out:  $3 + I_x = 5 + (-1) = 4 \Rightarrow I_x = 1 \text{ A (into the node)}$

(b) KCL on the boundary:  $2 + 7 = 4 + I_y \Rightarrow I_y = 5 \text{ A}$

### Trap

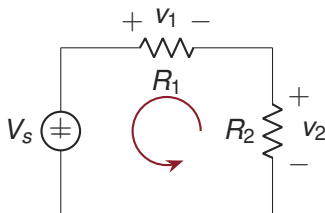
“−1 A out” is really 1 A *in*. Keep the sign as given and let the algebra do it.

### Engineering shortcut

In (b) you never needed to know what was inside the box.

# KVL: around any loop, voltages add to zero

[on formula sheet]



Walk the loop in one direction:

enter  $-$ , exit  $+$   $\Rightarrow$  **rise**: write  $+v$

enter  $+$ , exit  $-$   $\Rightarrow$  **drop**: write  $-v$

Clockwise here:  $+V_s - v_1 - v_2 = 0$

## Engineering shortcut

Across a resistor: walking *with* your current  $= -iR$ , walking *against* it  $= +iR$ .

# Assumed directions and negative answers

**You pick the direction. Being wrong is fine.**

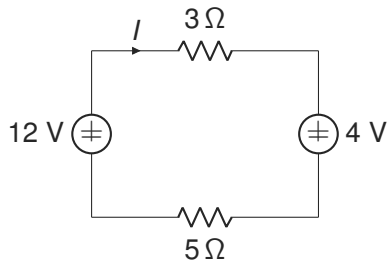
Before solving, draw an arrow for every unknown current, any direction you like. If the answer comes out negative, the real current flows the other way with that size.

**Common exam wording: “identify the direction of flow”**

Don't just write  $I = -2 \text{ A}$ . Write:  $I = -2 \text{ A}$ , so 2 A flows right to left .

## Try it: Example 7, one loop, two sources

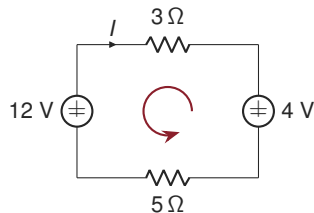
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Find  $I$  and the power of every element. Check the power balance.

your work here ↓

## Example 7: solution



**KVL (cw)**  $+12 - 3I - 4 - 5I = 0$

**Solve**  $8I = 8 \Rightarrow I = 1 \text{ A clockwise}$

**12 V**  $I$  leaves  $+$ : **supplies 12 W**

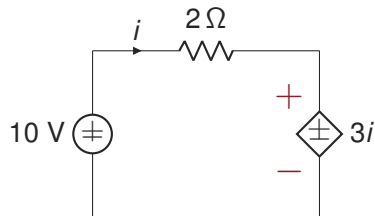
**4 V**  $I$  enters  $+$ : **absorbs 4 W**

**Resistors**  $3(1)^2 = 3 \text{ W}$ ,  $5(1)^2 = 5 \text{ W}$

**Check**  $12 = 4 + 3 + 5 \checkmark$

If you assumed counterclockwise, you'd get  $I = -1 \text{ A}$ . Same physical answer.

## Try it: Example 8, dependent source in a loop



The  $+$  of the  $3i$  source is on top.

Find  $i$  and the power of each element.

your work here ↓

## Example 8: solution

**KVL (cw)**  $+10 - 2i - 3i = 0$

**Solve**  $i = 2 \text{ A}$

**10 V**  $i$  leaves  $+$ : supplies  $10 \cdot 2 = 20 \text{ W}$

**$2 \Omega$**  absorbs  $2 \cdot 2^2 = 8 \text{ W}$

**$3i$  source**  $i$  enters its  $+$ :  $p = (3i)(i) = (6)(2) = 12 \text{ W absorbed}$

**Check**  $20 = 8 + 12 \checkmark$

Here the control variable  $i$  was already the unknown. If it were  $v_x$  elsewhere, add  $v_x = R \cdot (\text{its current})$  as a second equation.

# Series vs parallel: the definitions that work

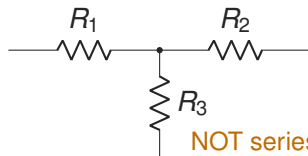
## The definitions

**Series:** share **one node**, and **nothing else** touches that node  $\Rightarrow$  same current.

**Parallel:** connected between the **same two nodes**  $\Rightarrow$  same voltage.



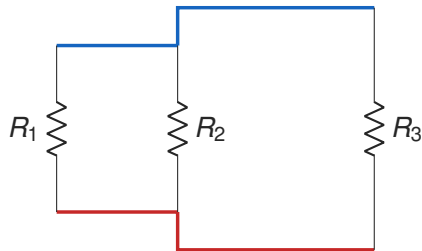
series ✓



NOT series!

$R_3$  touches the middle node, so the current can split there. This is the #1  $R_{eq}$  mistake.

# Spotting parallel: color the nodes



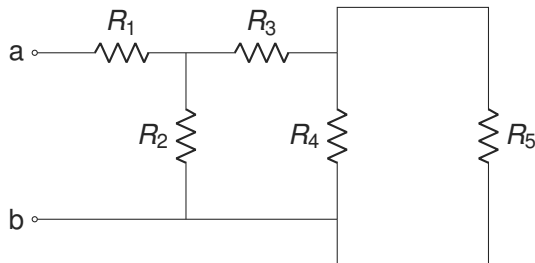
Every resistor touches **blue** and **red**.

Only 2 nodes, so all three are **in parallel**, even though  $R_3$  is drawn far away.

## Engineering shortcut: highlighter method

On the exam, trace each node in a different color (or number them). Two resistors with the *same pair of colors* are in parallel. The shape of the drawing means nothing, only the colors do.

## Try it: Example 9, spot series and parallel

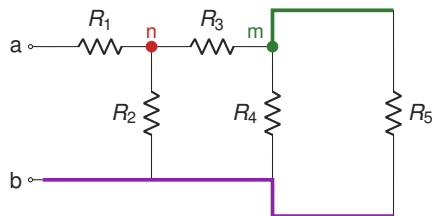


For each pair, say series, parallel, or neither:

- (a)  $R_4$  and  $R_5$
- (b)  $R_3$  and  $R_4$
- (c)  $R_1$  and  $R_2$
- (d)  $R_1$  and  $R_3$

your work here ↓

## Example 9: solution



- (a) parallel both go from m to b  
(b) neither  $R_5$  also touches m  
(c) neither  $R_3$  also touches n  
(d) neither  $R_2$  also touches n

After combining  $R_4 \parallel R_5$ , the result **is** in series with  $R_3$ . That's why you redraw after every step.

# Combining resistors

[on formula sheet]

Series:  $R_{eq} = R_1 + R_2 + \dots$       Parallel:  $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

## Parallel shortcuts (memorize)

Two resistors  $R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$  (only for TWO)

$N$  equal  $R$        $R/N$ :  $10 \parallel 10 = 5$ ,  $6 \parallel 6 \parallel 6 = 2$

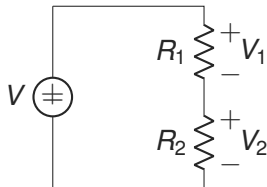
$R \parallel 2R = \frac{2}{3}R$ :  $1 \text{ k} \parallel 2 \text{ k} = 0.667 \text{ k}$

$R \parallel 0 = 0$  (a short kills everything across it)

**Sanity check:** parallel  $<$  smallest  $R$ , series  $>$  largest  $R$ .

# Voltage divider (series)

[on formula sheet]



$$V_2 = V \frac{R_2}{R_1 + R_2}$$

Same resistor on top. Any number of resistors:  $V_m = V \frac{R_m}{R_{eq}}$

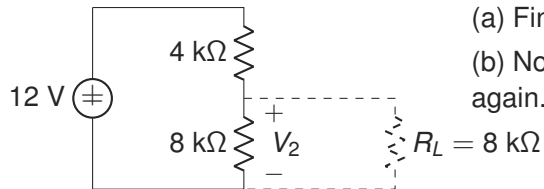
## Engineering shortcut

**Bigger  $R$  gets bigger  $V$ .** Equal resistors split it in half.  $R$  and  $2R$  split it 1/3 and 2/3.

## Trap

Only valid if nothing else connects to the middle node (Example 10).

## Try it: Example 10, the loaded divider trap



(a) Find  $V_2$  **without**  $R_L$ .

(b) Now connect  $R_L$  (dashed). Find  $V_2$  again.

your work here ↓

## Example 10: solution

(a) divider:  $V_2 = 12 \cdot \frac{8}{4 + 8} = 8 \text{ V}$

(b)  $R_L$  is in parallel with the 8 k:  $8 \parallel 8 = 4 \text{ k}$

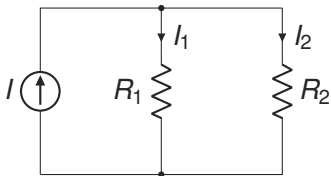
now divide against that:  $V_2 = 12 \cdot \frac{4}{4 + 4} = 6 \text{ V}$

### Trap

Adding a load **changes** the divider. Combine the load with the resistor it sits across first, then divide.

# Current divider (parallel)

[on formula sheet]



$$I_1 = I \frac{R_2}{R_1 + R_2}$$

**Other** resistor on top (two only).

Any number:  $I_m = I \frac{G_m}{G_{eq}}$

## The flip

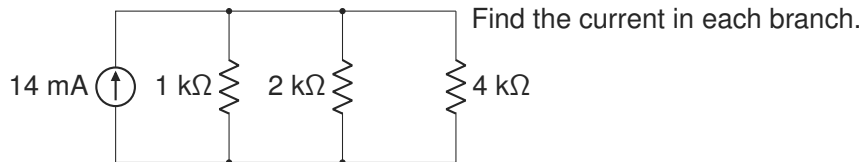
**Smaller  $R$  gets bigger  $I$** , the opposite of the voltage divider.

## Engineering shortcut

Often faster:  $V = I \cdot R_{eq}$  first, then  $V/R$  per branch.

## Try it: Example 11, current divider

---



your work here ↓

## Example 11: solution

**Conductances**  $G = 1 + 0.5 + 0.25 = 1.75 \text{ mS}$

**Shared voltage**  $V = I/G_{eq} = 14/1.75 = 8 \text{ V}$

**Each branch**  $8/1 = 8 \text{ mA}$ ,  $8/2 = 4 \text{ mA}$ ,  $8/4 = 2 \text{ mA}$

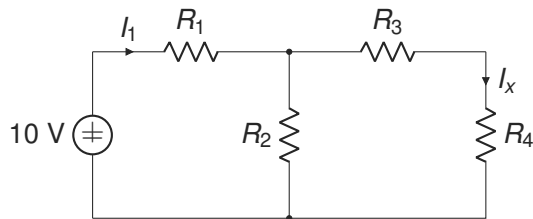
**Check (KCL)**  $8 + 4 + 2 = 14 \text{ mA}$  ✓

### Engineering shortcut

Voltage first, then Ohm's law per branch. There's no divider formula to flip by mistake, and it works for any number of branches.

## Try it: Example 12, ladder circuit

---



All  $R = 2 \text{ k}\Omega$ .

Find  $I_x$ .

your work here ↓

## Example 12: solution (collapse, then expand)

**1. Series**  $R_3 + R_4 = 4 \text{ k}$  (nothing else touches their shared node)

**2. Parallel**  $R_2 \parallel 4 \text{ k} = \frac{2 \cdot 4}{2+4} = \frac{4}{3} \text{ k}$

**3. Series** total  $= 2 + \frac{4}{3} = \frac{10}{3} \text{ k}$

**4. Source current**  $I_1 = 10 / \frac{10}{3} = 3 \text{ mA}$

**5. Divider**  $I_x = 3 \cdot \frac{2}{2+4} = I_x = 1 \text{ mA}$

**Check**  $V_{R_2} = (3 - 1)(2) = 4 \text{ V} = V_{R_3} + V_{R_4} = (1)(4) \checkmark$

# Ladder shortcut: assume $I_x = 1$

## Engineering shortcut: unit output method

Circuits are linear: **pretend**  $I_x = 1$ , walk back to the source, see what  $V_s$  you'd need, then scale.

**Assume**  $I_x = 1 \text{ mA} \Rightarrow V_{R_3+R_4} = (1)(4) = 4 \text{ V}$

$R_2$  same 4 V (parallel)  $\Rightarrow I_2 = 4/2 = 2 \text{ mA}$

**KCL**  $I_1 = 1 + 2 = 3 \text{ mA} \Rightarrow V_{R_1} = 6 \text{ V}$

**KVL** needed  $V_s = 6 + 4 = 10 \text{ V}$

**Scale** real  $V_s$  is also 10 V, so factor = 1  $\Rightarrow I_x = 1 \text{ mA}$

(Source of 20 V instead? Factor  $20/10 = 2$ , so  $I_x = 2 \text{ mA}$ .)

# Equivalent resistance: the method

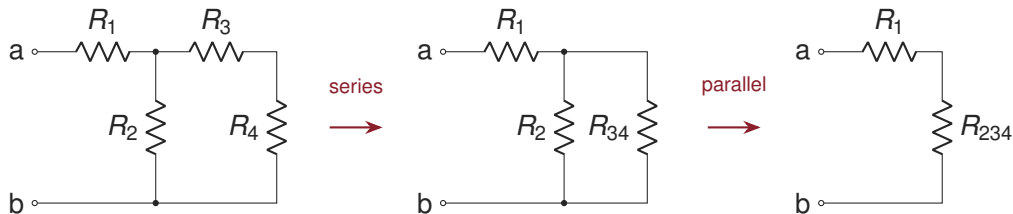
---

- 1 Color / label every node.
- 2 Start at the side **farthest** from the terminals a–b.
- 3 Combine one clear series or parallel pair.
- 4 **Redraw** the smaller circuit. Repeat.
- 5 Sanity check: parallel  $<$  smallest, series  $>$  largest.

## Engineering shortcut

Redraw after every combine. It feels slow, but it's where the marks are and it stops double-counting mistakes.

# The method in pictures

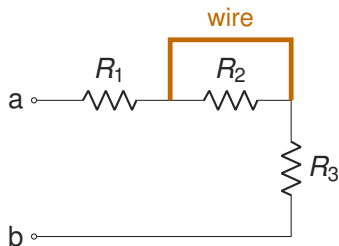


$$R_{34} = R_3 + R_4, \quad R_{234} = R_2 \parallel R_{34}, \quad \text{then } R_{ab} = R_1 + R_{234}.$$

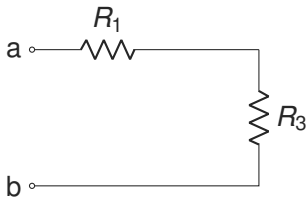
Notice each drawing is simpler than the last. Never skip a drawing on the exam.

## $R_{eq}$ trap 1: a shorted resistor disappears

Drawn like this



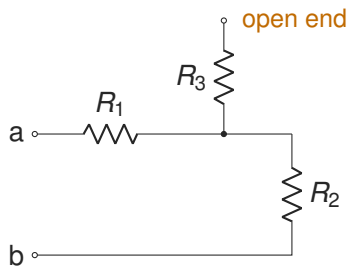
Is really this



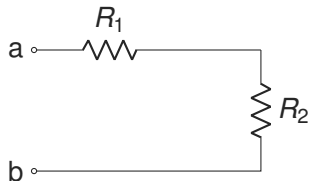
The wire connects both ends of  $R_2$ , so  $V_{R_2} = 0$  and no current goes through it. Delete  $R_2$ :  $R_{ab} = R_1 + R_3$ .

## $R_{eq}$ trap 2: a dangling resistor

Drawn like this

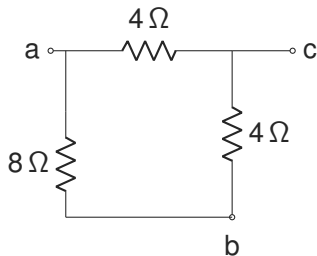


Is really this



One end of  $R_3$  goes nowhere, so no current can flow through it. Delete it. It does **not** add to the chain:  $R_{ab} = R_1 + R_2$ .

## $R_{eq}$ trap 3: the terminals matter



Same 3 resistors, 3 different questions:

$$R_{ab} \quad 8 \parallel (4 + 4) = 4 \, \Omega$$

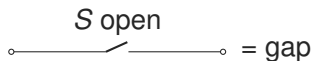
$$R_{ac} \quad 4 \parallel (8 + 4) = 3 \, \Omega$$

$$R_{cb} \quad 4 \parallel (4 + 8) = 3 \, \Omega$$

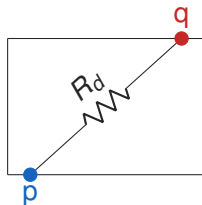
### Engineering shortcut

Put your finger on the two terminals. The resistor directly between them is in parallel with “everything else” between them.

## $R_{eq}$ trap 4: switches and diagonals



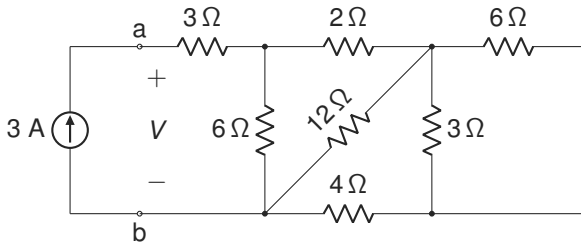
**Switches:** redraw the circuit once for **each** switch case. Exams love this.



**Diagonal resistors** are nothing special. Find the two nodes they connect (here  $p$  and  $q$ ) and treat it like any other resistor.

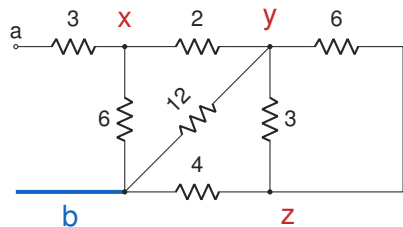
## Try it: Example 13, exam-style $R_{eq}$

(a) Find  $R_{eq}$  to the right of a–b. (b) Use it to find  $V$ .



your work here ↓

## Example 13: solution (1/2)



**y-z**  $3 \parallel 6 = 2$

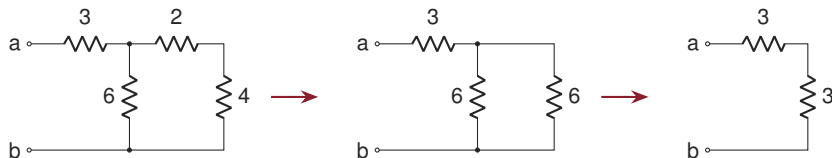
**y-b**  $2 + 4 = 6$  (only 2 things touch z)

**y-b** diagonal  $12 \parallel 6 = 4$

### Engineering shortcut

The whole blue bottom-left wire is node b, including the bottom of the 6  $\Omega$  and the low end of the diagonal 12  $\Omega$ . Seeing that is the whole question.

## Example 13: solution (2/2)

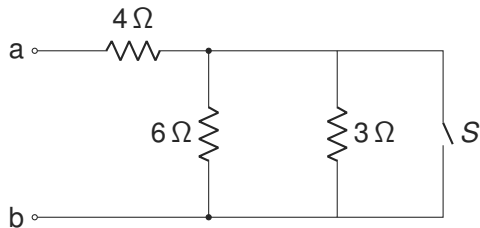


**x-b**  $2 + 4 = 6$ , then  $6 \parallel 6 = 3$

**a-b**  $3 + 3 = R_{eq} = 6 \Omega$

**(b)**  $V = (3 \text{ A})(6 \Omega) = V = 18 \text{ V}$

## Try it: Example 14, switch



Find  $R_{ab}$ :

- (a)  $S$  open
- (b)  $S$  closed

your work here ↓

## Example 14: solution

(a) **open** switch = gap, ignore it:  $6 \parallel 3 = 2$ ,  $4 + 2 = 6 \Omega$

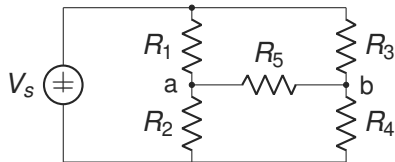
(b) **closed** switch = wire between the same two nodes as the 6 and the 3

$$6 \parallel 3 \parallel 0 = 0, \text{ so } 4 + 0 = 4 \Omega$$

### Trap

A closed switch shorts **everything** in parallel with it, not just the resistor drawn next to it.

# Wheatstone bridge



$R_5$  connects the middle of both sides.  
It's **not** series or parallel with anything, so the usual combining fails.

**Balanced** when

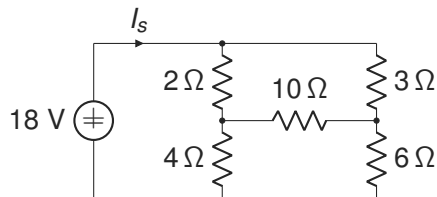
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Then  $v_a = v_b$ , so **no current** flows in  $R_5$ .

## Engineering shortcut: balanced bridge

Check the ratio first. If it's balanced, delete  $R_5$  (or short it, same answer). What's left is simple series/parallel.

## Try it: Example 15, bridge



Find the current in the  $10\ \Omega$ , the equivalent resistance seen by the source, and  $I_s$ .

your work here ↓

## Example 15: solution

**Ratio**  $2/4 = 0.5$  and  $3/6 = 0.5 \Rightarrow$  balanced

$10\ \Omega$   $I_{10} = 0$ , so delete it

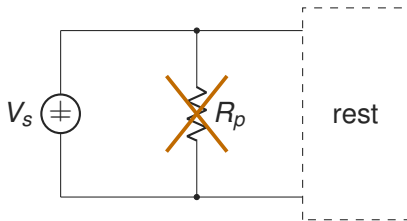
**Left side**  $2 + 4 = 6\ \Omega$     **Right side:**  $3 + 6 = 9\ \Omega$

$R_{eq}$   $6 \parallel 9 = 54/15 = 3.6\ \Omega$

$I_s$   $18/3.6 = 5\ \text{A}$

**Check**  $V_a = 18 \times 4/6 = 12\ \text{V}$ ,  $V_b = 18 \times 6/9 = 12\ \text{V}$ , equal ✓

# Cross-out trap 1: $R$ across a $V$ source

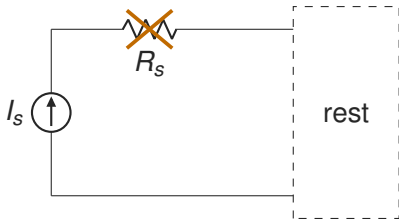


$R_p$  always has exactly  $V_s$  across it, no matter what the rest does.

So for the **rest of the circuit**,  $R_p$  changes nothing. Cross it out.

It only changes the **source's own current**:  $I_{\text{source}} = I_{\text{rest}} + V_s/R_p$ .

## Cross-out trap 2: $R$ in series with an $I$ source

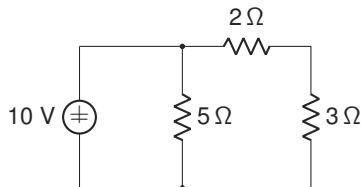


$R_s$  always carries exactly  $I_s$ , no matter what the rest does.

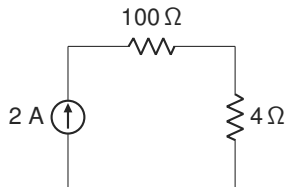
So for the **rest of the circuit**,  $R_s$  acts like a wire. Cross it out.

It only changes the **source's own voltage**:  $V_{\text{source}} = V_{\text{rest}} + I_s R_s$ .

## Try it: Example 16, cross out



(a)



(b)

(a) Current in the  $3\ \Omega$ , and the source current.  
and across the current source.

(b) Voltage across the  $4\ \Omega$ ,

your work here ↓

## Example 16: solution

- (a)  $5\ \Omega$  is across the  $10\ \text{V}$  source: cross it out for the rest

$$I_{3\Omega} = 10/(2 + 3) = 2\ \text{A}$$

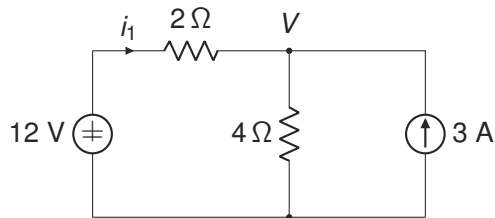
$5\ \Omega$  draws  $10/5 = 2\ \text{A}$ , so the source gives  $2 + 2 = 4\ \text{A}$

- (b)  $100\ \Omega$  is in series with the  $2\ \text{A}$  source: cross it out for the rest

$$V_{4\Omega} = 2 \times 4 = 8\ \text{V} \quad (\text{the } 100\ \Omega \text{ changes nothing here})$$

$$\text{source voltage} = 2 \times (100 + 4) = 208\ \text{V} \quad (\text{top } +)$$

## Try it: Example 17, both kinds of source



Bottom = 0 V.

Find  $V$  and the power of each source.

your work here ↓

## Example 17: solution

**KCL at  $V$**  out = in:  $\frac{V}{4} = \frac{12 - V}{2} + 3$

**Solve**  $\times 4$ :  $V = 24 - 2V + 12 \Rightarrow V = 12 \text{ V}$

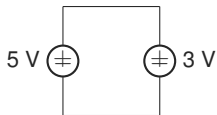
$i_1$   $(12 - 12)/2 = 0$ : the 12 V source gives **zero** power

**3 A source** pushes 3 A out at +12 V: supplies 36 W

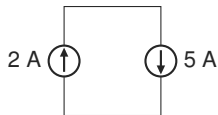
**Check**  $4 \Omega$  absorbs  $12^2/4 = 36 \text{ W}$  ✓

A voltage source's current is whatever the circuit decides, including zero.  
(This is a preview of nodal analysis, week 2.)

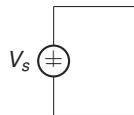
# Impossible circuits



different  $V$  sources in parallel: **breaks KVL**



different  $I$  sources in series: **breaks KCL**



wire across a  $V$  source: **infinite current**

If you think you see one of these on an exam, you misread the circuit. (A wire across a *current* source is fine: all the current takes the wire.)

# Sign and formula traps

## Check these before you box an answer

- $v = iR$  needs  $i$  entering the  $+$  side. Otherwise  $v = -iR$ .
- Product-over-sum is for **two** resistors only.
- The current divider uses the **other**  $R$  on top. The voltage divider uses the **same**  $R$ .
- Resistor power is always  $\geq 0$ . Source power can be either sign.
- Every ground symbol is the same node.
- A negative answer is fine: state the real direction.

# Exam-format traps

---

**Using a different method than the one requested can get 0 marks**

If they say nodal, use nodal, even if something else is faster.

**“First give the equations with variables, then replace with values”**

Write the symbolic equation first. It's worth marks by itself.

**Box your final answer**

Include units, and a direction for currents.

# Summary (1/2)

---

## Basics

$$i = dq/dt, \quad p = vi, \quad W = Pt, \quad 1 \text{ A}\cdot\text{h} = 3600 \text{ C}, \quad R = \rho\ell/A$$

## Ohm

$$v = iR \text{ (} i \text{ into +)}, \quad p = vi = i^2 R = v^2/R$$

## PSC

$$i \text{ into +: } p = vi. \quad p > 0 \text{ absorbs, } p < 0 \text{ supplies. } \sum p = 0$$

## KCL / KVL

$$\sum i_{in} = \sum i_{out} \text{ (also on regions)} \quad / \quad \sum v = 0 \text{ (rise +, drop -)}$$

## Counting

$$\text{meshes} = b - n + 1 \quad \bullet \quad \text{one wire} = \text{one node}$$

## Series

$$\text{one node, nothing else attached, same } i, \quad R_{eq} = \sum R$$

## Parallel

$$\text{same two nodes, same } v, \text{ two: } \frac{R_1 R_2}{R_1 + R_2}$$

# Summary (2/2)

---

|                  |                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dividers</b>  | $V_m = V \frac{R_m}{R_{eq}}$ (same $R$ ) • $I_1 = I \frac{R_2}{R_1 + R_2}$ (other $R$ ) • a load changes the divider                           |
| <b>Sources</b>   | $V$ fixes $v$ (its $i$ is free), $I$ fixes $i$ (its $v$ is free). Killed: $V \rightarrow$ short, $I \rightarrow$ open. Dependent: never killed |
| $R_{eq}$         | color nodes, far end first, redraw each step. Delete shorted and dangling $R$ . Terminals matter. Balanced bridge: delete the middle           |
| <b>Cross out</b> | $R$ across a $V$ source, and $R$ in series with an $I$ source                                                                                  |
| <b>Shortcuts</b> | $k\Omega \cdot mA = V$ • $R/N$ • voltage first, then $V/R$ • ladder: assume $I_x = 1$                                                          |
| <b>Checks</b>    | KCL, KVL, power balance, parallel $<$ smallest, series $>$ largest                                                                             |

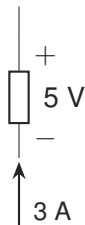
# Final check

8 problems. One per slide, with space. No looking back.

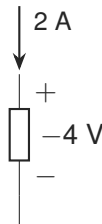
If one takes more than 10 min, go back to that section. The answer sheet comes after P8.

# P1: sign convention

---



(a)



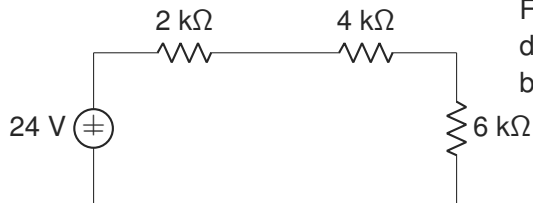
(b)

Power absorbed by each? Absorbing or supplying?

your work here ↓

## P2: series circuit

---

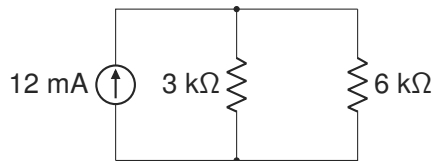


Find the current,  $V_{4k}$  (use the divider), and the power supplied by the source.

your work here ↓

## P3: current divider

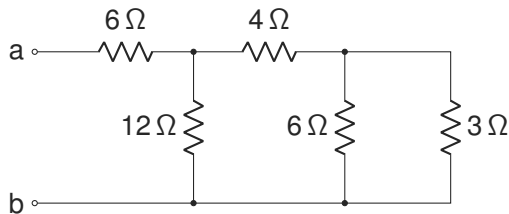
---



Find each branch current and the voltage across them.

your work here ↓

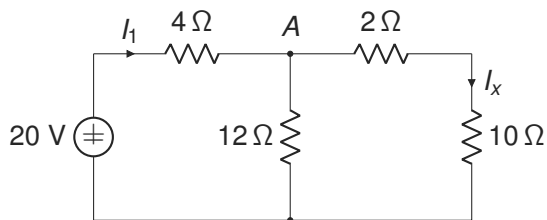
## P4: equivalent resistance



Find  $R_{ab}$ .

your work here ↓

## P5: ladder



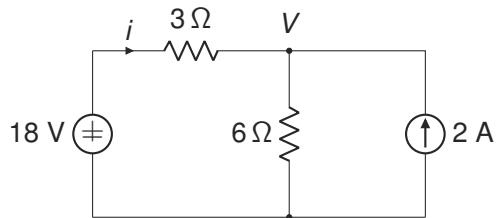
Find  $I_1$ ,  $V_A$ ,  $I_x$ .

(Try the “assume  $I_x = 1$ ”  
trick.)

your work here ↓

## P6: two sources

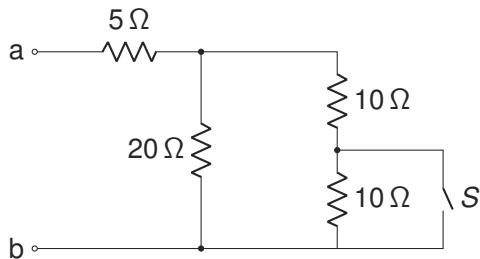
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Find  $V$  and the power supplied by each source. Check the balance.

your work here ↓

## P7: switch

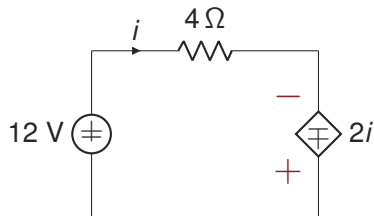


Find  $R_{ab}$  with  $S$  open and with  $S$  closed.

your work here ↓

## P8: dependent source that helps

---



The  $+$  of the  $2i$  source is at the **bottom**.

Find  $i$ . Does the dependent source absorb or supply, and how much?

your work here ↓

# Answer sheet

Full worked solutions to P1–P8.

# P1: solution

---

(a) 3 A enters the **bottom** (−), so it leaves +

$$p = -vi = -(5)(3) = -15 \text{ W} \Rightarrow \text{supplies 15 W}$$

(b) 2 A enters +  $\Rightarrow p = vi = (-4)(2) = -8 \text{ W}$

$$\Rightarrow \text{supplies 8 W}$$

## P2: solution

---

**Total  $R$**   $2 + 4 + 6 = 12 \text{ k}\Omega$

**Current**  $I = 24/12 = 2 \text{ mA}$  ( $\text{V/k}\Omega = \text{mA}$ )

**Divider**  $V_{4k} = 24 \cdot \frac{4}{12} = 8 \text{ V}$  (check:  $2 \text{ mA} \times 4 \text{ k} = 8 \text{ V}$  ✓)

**Power**  $P = 24 \times 2 \text{ mA} = 48 \text{ mW supplied}$

**Check**  $I^2 R: 8 + 16 + 24 = 48 \text{ mW}$  ✓

## P3: solution

---

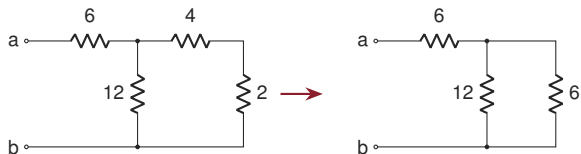
$$R_{eq} \quad 3 \parallel 6 = \frac{18}{9} = 2 \text{ k}\Omega$$

$$\text{Voltage} \quad V = 12 \text{ mA} \times 2 \text{ k} = 24 \text{ V}$$

$$\text{Branches} \quad 24/3 = 8 \text{ mA}, \quad 24/6 = 4 \text{ mA}$$

$$\text{Check} \quad 8 + 4 = 12 \checkmark \quad \text{smaller } R \text{ got more current } \checkmark$$

## P4: solution



**Far end**  $6 \parallel 3 = 2$

**Series**  $4 + 2 = 6$

**Parallel**  $12 \parallel 6 = 4$

**Final**  $6 + 4 = R_{ab} = 10 \Omega$

## P5: solution (two ways)

### Collapse

1  $2 + 10 = 12$

2  $12 \parallel 12 = 6$

3  $4 + 6 = 10 \Omega$

4  $I_1 = 20/10 = 2 \text{ A}$

5  $V_A = 20 - 4 \cdot 2 = 12 \text{ V}$

6  $I_x = 12/12 = 1 \text{ A}$

### Assume $I_x = 1 \text{ A}$

1  $V_A = 1 \cdot (2 + 10) = 12 \text{ V}$

2  $I_{12} = 12/12 = 1 \text{ A}$

3  $I_1 = 1 + 1 = 2 \text{ A}$

4  $V_s = 12 + 4 \cdot 2 = 20 \text{ V}$

5 matches, so factor = 1 ✓

## P6: solution

**KCL at V**  $\frac{V}{6} = \frac{18 - V}{3} + 2$

**Solve**  $\times 6: V = 36 - 2V + 12 \Rightarrow V = 16 \text{ V}$

**i**  $(18 - 16)/3 = 2/3 \text{ A}$

**18 V source**  $18 \cdot \frac{2}{3} = 12 \text{ W supplied}$

**2 A source**  $2 \cdot 16 = 32 \text{ W supplied}$

**Absorbed**  $6 \Omega: 16^2/6 = 42.67 \text{ W}, \quad 3 \Omega: (2/3)^2 \cdot 3 = 1.33 \text{ W}$

**Check**  $12 + 32 = 44 = 42.67 + 1.33 \quad \checkmark$

## P7: solution

**Open** right branch =  $10 + 10 = 20$ ,  $20 \parallel 20 = 10$

$$R_{ab} = 5 + 10 = 15 \Omega$$

**Closed** S shorts the lower 10, so the right branch = 10

$$20 \parallel 10 = \frac{200}{30} = 6.67, R_{ab} = 5 + 6.67 = 11.67 \Omega$$

### Trap

Here the switch is across *one* resistor only, so it removes only that one. Compare with Example 14, where it was across the whole parallel group.

## P8: solution

**KVL (cw)** up the 12 V:  $+12$ ,  $4\ \Omega$ :  $-4i$ , down through  $2i$  entering its  $-$ :  $+2i$

**Equation**  $12 - 4i + 2i = 0 \Rightarrow i = 6\text{ A}$

**Dep. source**  $i$  enters its  $-$ :  $p = -(2i)(i) = -72\text{ W} \Rightarrow$   
supplies 72 W

**Check** 12 V supplies 72 W, resistor absorbs  $4 \cdot 36 = 144\text{ W}$ :  
 $72 + 72 = 144\ \checkmark$

Dependent sources can supply power too. Only the sign decides.